

Moein Razavi

AI/ML Engineer | Data Scientist | Software Engineer

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Professional Summary

Data scientist and software developer with a Ph.D. in Industrial Engineering. Develops AI applications, forecasting and energy decision systems at Ford, with additional experience in wearable health software. Builds data pipelines, machine learning models, cloud APIs, and mobile and web interfaces. Research has 500+ citations on Google Scholar as of September 2026.

Professional Experience

Ford Motor Company

Data Scientist | Aug 2023 - Present

- **Utility data onboarding (HOMeload):** Built a Python/FastAPI pipeline standardizing customer electricity records for energy analysis. Combined Vertex AI Gemini schema discovery with pandas/lxml extraction, deterministic schema/timestamp/range checks, and bounded corrective retries. Used Cloud Storage and BigQuery MERGE for repeatable loads; deployed with Docker, Cloud Run, and Terraform.
- **EV analytics assistant:** Built a LangChain multi-agent assistant using GPT-4o/GPT-4o-mini and rules/LLM routing to BM25-based retrieval-augmented generation (RAG) for legal documents and vehicle manuals, or schema-aware text-to-SQL over BigQuery. Added conversational context and LLM review against domain rules; deployed a Dash interface on Cloud Run with SAML/ADFS authentication.
- **Vehicle-to-grid market bidding:** Built a Python/PyTorch system using vehicle-availability forecasts to inform fleet energy bids. Selected PatchTST as the best-performing model in project evaluations; tuned candidates with Optuna on Vertex AI GPUs. Combined ERCOT prices via gridstatus with Monte Carlo capacity estimates, greedy vehicle selection, and load reallocation after vehicle disconnections to support bidding under uncertainty.
- **Rate recommendation:** Developed XGBoost electricity-bill predictions using utility tariffs, vehicle telemetry, and household consumption to inform utility-plan recommendations. Built Flask APIs, BigQuery workflows, and a Dash interface with SAML authentication; deployed on Cloud Run through Tekton CI/CD.
- **Energy data engineering and reporting:** Built PySpark/BigQuery pipelines and Airflow DAGs for energy-simulator inputs and optimizer outputs. Developed discharge reporting in Looker Studio and REST APIs for operational dashboards.
- **Code analysis and cloud cost estimation:** Built a FastAPI/LangChain tool using GitPython and Python AST parsing to extract functions and loops for LLM-assisted complexity estimates. Streamed findings through Server-Sent Events; estimated Vertex AI compute costs from model-generated resource assumptions and a static pricing table.

CereVu Medical, Inc.

Software Developer / Research Engineer (Contract) | Jun 2023 - Sep 2024

- **Wearable health platform:** Developed Vitality and Clinical in SwiftUI, using CoreBluetooth BLE and Swift Charts to acquire and display 9 physiological signals. Migrated Clinical from UIKit to MVVM-Cordinator; implemented role-based workflows, calibration, exports, and OAuth 2.0/JWT authentication.
- **Backend and delivery:** Built Python/Flask services with WebSocket biosignal streaming and a React/Next.js dashboard backed by MongoDB. Deployed Docker containers on AWS EC2 with GitHub Actions and distributed iOS builds through TestFlight.

Education

Ph.D., Industrial Engineering | Texas A&M University | Dec 2023

Master of Engineering, Computer Engineering | Texas A&M University | May 2021

Selected Projects

- **Business Action Prioritizer:** Built a Google ADK/Gemini agent workflow that converts uploaded data and a scoring rubric into ranked actions. Required user approval before executing plans with PandasAI on local dataframes; used FastAPI/SQLite for sessions and an MCP server to publish reports to Cloud Storage.
- **InspecaLytics:** Built SwiftUI and React/TypeScript inspection apps with Supabase, annotated photo capture, and Word/PowerPoint reports. Implemented offline synchronization with field-level conflict resolution and XCTest/Vitest coverage. Combined ARKit spatial capture with Apple Vision image registration and perceptual similarity to organize inspection photos on-device.
- **Clink Social:** Built SwiftUI messaging and MapKit discovery with Firebase Auth/Firestore/FCM. Implemented feed aggregation and notifications in TypeScript Cloud Functions, plus QR-based cross-device sign-in through time-limited OAuth/JWT token exchange.

Research Experience

Texas A&M University

Research Assistant | Sep 2018 - May 2024

- **Wearable stress detection:** Engineered windowed heart-rate/motion features in Python from 54 students over 40 days, with self-reported stress labels. Compared classifiers including XGBoost using cross-validation and ROC-AUC; interpreted predictions with SHAP.
- **MERCURY:** Co-authored research using random projection with quantization and cached computation reuse; reported 1.97x average training speedup across 12 models with accuracy similar to baseline.
- **OpenSync:** Co-developed open-source Python software using Lab Streaming Layer to synchronize sensors and experiment events; published tests measured RMS recording-time lag below 0.07 ms.
- **Computer vision:** Developed face-mask and physical-distance monitoring using TensorFlow Faster R-CNN detectors and calibrated perspective transforms to measure distances in video.
- **GlucoseCoach:** Built an iOS app and Python/Flask backend for personalized glucose monitoring.

Technical Skills

Languages: Python, SQL, Swift, TypeScript, JavaScript, C++, Java, R, MATLAB

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, Optuna, OpenCV, PatchTST, forecasting, Monte Carlo simulation, deep learning, cross-validation, SHAP/LIME

Generative AI: LangChain, Google ADK, Gemini, RAG, BM25, multi-agent routing, prompt engineering, schema-aware text-to-SQL, human-in-the-loop workflows, MCP, PandasAI

Data: pandas, NumPy, SciPy, PySpark, Airflow, ETL/ELT, PostgreSQL, MongoDB, Redis, Firebase, Supabase

Applications: FastAPI, Flask, REST APIs, WebSocket, React, Next.js, SwiftUI, UIKit, CoreBluetooth, ARKit, Apple Vision, MVVM, OAuth 2.0, JWT, SAML

Cloud: AWS (S3, EC2, Lambda, SageMaker), GCP (Vertex AI, BigQuery, Cloud Run, Dataproc), Azure OpenAI

Delivery: Git, Docker, Kubernetes, Terraform, CI/CD, Tekton, GitHub Actions, AWS CodeDeploy, Fastlane, TestFlight, XCTest, Vitest

Selected Publications

Razavi, M., et al. Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review. JMIR Mental Health, 2024.

Janfaza, V., Weston, K., Razavi, M., et al. MERCURY: Accelerating DNN Training By Exploiting Input Similarity. IEEE HPCA, 2023.

Razavi, M., et al. OpenSync: An open-source platform for synchronizing multiple measures in neuroscience experiments. Journal of Neuroscience Methods, 2022.

Razavi, M., et al. An Automatic System to Monitor the Physical Distance and Face Mask Wearing of Construction Workers in COVID-19 Pandemic. SN Computer Science, 2022.